07 - Model Training and Hyperparameter Tuning

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Training a Decision Tree

```
model_dt.fit(x_train, y_train)
y_pred = model_dt.predict(x_test)
```

fit() learns the tree structure from labelled training data. predict() applies the learned rules to new feature rows.

2. SMOTE: synthetic minority oversampling

SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic minority-class observations using interpolation between an observation and one of its minority-class nearest neighbours.

$$x_new = x_i + \lambda(x_zi - x_i), \text{ where } \lambda \text{ is sampled between 0 and 1}$$

This creates points along the segment joining two minority-class observations. It differs from random oversampling, which duplicates existing minority examples. SMOTE can still create overlapping or unrealistic examples, especially when features are categorical or the minority class has complex structure.

3. ENN and SMOTEENN

Edited Nearest Neighbours (ENN) is a cleaning/undersampling method that removes observations whose class labels are inconsistent with their nearest neighbours. SMOTEENN first applies SMOTE and then ENN cleaning. The intention is to increase minority representation and remove ambiguous/noisy examples.

Important: The repository uses SMOTEENN on the complete dataset before the train-test split. This is a methodological error. Use the resampler only inside the training portion of an imbalanced-learn Pipeline. Also consider categorical-aware resampling such as SMOTENC where appropriate; plain SMOTE on one-hot encoded categorical variables can create fractional dummy values.

4. Random Forest training

```
model_rf_smote = RandomForestClassifier(
   n_estimators=100, criterion="gini",
   random_state=100, max_depth=6,
   min_samples_leaf=8
)
model_rf_smote.fit(xr_train1, yr_train1)
```

The model is fit using the resampled training partition as generated by the notebook's workflow. Its reported evaluation must be interpreted with the leakage caveat described in the validation section.

5. Was hyperparameter tuning performed?

The repository specifies hyperparameters manually. It does not demonstrate GridSearchCV, RandomizedSearchCV, Bayesian optimization, or a systematic cross-validated tuning procedure. Do not claim that formal hyperparameter tuning was performed.

Suggested tuning approach

```
from imblearn.pipeline import Pipeline
from imblearn.combine import SMOTEENN
from sklearn.model_selection import RandomizedSearchCV, StratifiedKFold
from sklearn.ensemble import RandomForestClassifier


pipe = Pipeline([
   ("resample", SMOTEENN(random_state=42)),
   ("model", RandomForestClassifier(random_state=100))
])
param_dist = {
   "model__n_estimators": [100, 200, 300],
   "model__max_depth": [4, 6, 8, 10, None],
   "model__min_samples_leaf": [2, 4, 8, 10],
   "model__max_features": ["sqrt", "log2"]
}
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
search = RandomizedSearchCV(
   pipe, param_distributions=param_dist,
   n_iter=25, scoring="average_precision",
   cv=cv, random_state=42, n_jobs=-1
)
search.fit(X_train, y_train)
```

The exact scoring metric should reflect the business objective. Average precision / PR-AUC can be informative for the positive minority class; recall at a specified precision or a cost-based objective may be more appropriate in some settings.

6. Interview-ready explanation

> *The notebooks train Decision Tree and Random Forest classifiers and use SMOTEENN to address class imbalance. SMOTE creates synthetic minority observations, while ENN removes observations based on nearest-neighbour disagreement. The hyperparameters were manually specified; there is no documented GridSearchCV or RandomizedSearchCV. A stronger approach is to place resampling and the classifier in a pipeline and tune it using stratified cross-validation.*

7. Questions to prepare

Q. How does SMOTE generate a sample?

It interpolates between a minority observation and one of its minority-class nearest neighbours.

Q. What does ENN do?

It removes observations considered inconsistent with their nearest-neighbour labels.

Q. Why is resampling inside cross-validation?

To ensure each fold's validation data does not influence synthetic sample generation or cleaning.

Q. Grid search versus random search?

Grid search evaluates every specified combination. Random search samples a fixed number of combinations and can explore broad spaces more efficiently when only a few hyperparameters matter strongly.

Q. What is overfitting?

A model fits training-specific patterns or noise and performs substantially worse on unseen data.